

IIE6120 Linear Programming
2025-2 Homework 02
Team B (이정민, 장서현)

#1

[2.16] If the i th row of a square nonsingular matrix $\mathbf{B}$ is multiplied by a scalar $\lambda \neq 0$, what changes would result in $\mathbf{B}^{-1}$?

Let $\mathbf{B} \in \mathbb{R}^{n \times n}$ and $\tilde{\mathbf{B}} \in \mathbb{R}^{n \times n}$ where the i th row of $\mathbf{B}$ is multiplied by scalar $\lambda \neq 0$.

$$\tilde{\mathbf{B}} = \mathbf{E}_i(\lambda) \mathbf{B} \text{ where } \mathbf{E}_i(\lambda) = \text{diag}(1, \dots, 1, \lambda, 1, \dots, 1)$$

$$\tilde{\mathbf{B}}^{-1} = [\mathbf{E}_i(\lambda) \mathbf{B}]^{-1} = \mathbf{B}^{-1} \mathbf{E}_i(\lambda)^{-1}$$

$$\text{Since } \mathbf{E}_i(\lambda) \text{ is diagonal, } \mathbf{E}_i(\lambda)^{-1} = \text{diag}(1, \dots, 1, \frac{1}{\lambda}, 1, \dots, 1)$$

$$\text{Thus } \tilde{\mathbf{B}}^{-1} = \mathbf{B}^{-1} \text{diag}(1, \dots, 1, \frac{1}{\lambda}, 1, \dots, 1)$$

$\therefore$ Only the i th column of $\mathbf{B}^{-1}$ is multiplied by $1/\lambda$. All the other columns remain unchanged.

#2

[2.17] If the i th column of a square nonsingular matrix $\mathbf{B}$ is multiplied by a scalar $\lambda \neq 0$, what changes would result in $\mathbf{B}^{-1}$?

Let $\mathbf{B} \in \mathbb{R}^{n \times n}$ and $\tilde{\mathbf{B}} \in \mathbb{R}^{n \times n}$ where the i th col. of $\mathbf{B}$ is multiplied by scalar $\lambda \neq 0$.

$$\tilde{\mathbf{B}} = \mathbf{B} \mathbf{E}_i(\lambda) \text{ where } \mathbf{E}_i(\lambda) = \text{diag}(1, \dots, 1, \lambda, 1, \dots, 1)$$

$$\tilde{\mathbf{B}}^{-1} = [\mathbf{B} \mathbf{E}_i(\lambda)]^{-1} = \mathbf{E}_i(\lambda)^{-1} \mathbf{B}^{-1}$$

$$\text{Since } \mathbf{E}_i(\lambda) \text{ is diagonal, } \mathbf{E}_i(\lambda)^{-1} = \text{diag}(1, \dots, 1, \frac{1}{\lambda}, 1, \dots, 1)$$

$$\text{Thus } \tilde{\mathbf{B}}^{-1} = \text{diag}(1, \dots, 1, \frac{1}{\lambda}, 1, \dots, 1) \mathbf{B}^{-1}$$

$\therefore$ Only the i th row of $\mathbf{B}^{-1}$ is multiplied by $1/\lambda$. All the other rows remain unchanged.

#3

[2.18] Find the rank of the following matrices:

$$(i) \mathbf{A} = \begin{bmatrix} 1 & 0 & 1 & -1 \\ 2 & 3 & 4 & 1 \\ 1 & 0 & 5 & -3 \end{bmatrix}$$

$$(ii) \mathbf{A} = \begin{bmatrix} -2 & 1 & -3 \\ 2 & 4 & -2 \\ 4 & 3 & 1 \end{bmatrix}$$

$$(i) R_2' \leftarrow 2R_1 - R_2$$

$$2R_1 : 2 \ 0 \ 2 \ -2$$

$$\rightarrow R_2 : 2 \ 3 \ 4 \ 1$$

$$R_2' : 0 \ -3 \ -2 \ -3$$

$$R_2'' : 0 \ 1 \ 2/3 \ 1$$

$$R_3' \leftarrow R_1 - R_3$$

$$R_1 : 1 \ 0 \ 1 \ -1$$

$$\rightarrow R_3 : 1 \ 0 \ 5 \ -3$$

$$R_3' : 0 \ 0 \ 4 \ 2$$

$$R_3'' : 0 \ 0 \ 1 \ -1/2$$

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 2/3 & 1 \\ 1 & 0 & 5 & -3 \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 2/3 & 1 \\ 0 & 0 & 1 & -1/2 \end{bmatrix}$$

$$\therefore \text{rank}(\mathbf{A}) = 3$$

$$(ii) R_2' \leftarrow R_1 + R_2$$

$$R_1 : -2 \ 1 \ -3$$

$$+ R_2 : 2 \ 4 \ -2$$

$$R_2' : 0 \ 5 \ -5$$

$$R_2'' : 0 \ 1 \ -1$$

$$R_3' \leftarrow 2R_1 + R_3$$

$$2R_1 : -4 \ 2 \ -6$$

$$+ R_3 : 4 \ 3 \ 1$$

$$R_3' : 0 \ 5 \ -5$$

$$R_3'' : 0 \ 1 \ -1$$

$$\mathbf{A} = \begin{bmatrix} -2 & 1 & -3 \\ 0 & 1 & -1 \\ 4 & 3 & 1 \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} -2 & 1 & -3 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$R_3'' \leftarrow R_2 - R_3$$

$$R_1' \leftarrow -\frac{1}{2} R_1$$

$$\mathbf{A} = \begin{bmatrix} 1 & -1/2 & 3/2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore \text{rank}(\mathbf{A}) = 2$$

#4

[2.21] Demonstrate by enumeration that every basis matrix of the following system is triangular:

$$\begin{aligned} x_1 + x_3 + x_4 &= 4 \\ -x_1 + x_2 + x_5 &= 2 \\ -x_2 - x_4 &= -3 \\ -x_3 &= -2. \end{aligned}$$

$$A = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ -1 & 1 & 0 & 0 & 1 \\ 0 & -1 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 2 \\ -3 \\ -2 \end{bmatrix}$$

$$= [a_1 \ a_2 \ a_3 \ a_4 \ a_5]$$

Basis $B \in \mathbb{R}^{4 \times 4}$.

a_3 must be included

and since a_1, a_2, a_4 are linearly dependent,
($\because a_4 = a_1 + a_2$)

a_5 must also be included.

① exclude a_1

$$B = [a_2 \ a_3 \ a_4 \ a_5] \sim [a_3 \ a_4 \ a_2 \ a_5]$$

$$= \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & -1 & -1 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} -1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

:triangular

② exclude a_2

$$B = [a_1 \ a_3 \ a_4 \ a_5] \sim [a_4 \ a_3 \ a_1 \ a_5]$$

$$= \begin{bmatrix} 1 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

:triangular

③ exclude a_4

$$B = [a_1 \ a_2 \ a_3 \ a_5] \sim [a_2 \ a_3 \ a_1 \ a_5]$$

$$= \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 1 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & -1 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & -1 & 1 \end{bmatrix}$$

:triangular

#5 [2.24] Find all basic solutions of the following system:

$$\begin{aligned} -x_1 + 2x_2 + x_3 + 3x_4 - 2x_5 &= 4 \\ x_1 - 2x_2 + 2x_4 + x_5 &= 2. \end{aligned}$$

$$A = \begin{bmatrix} -1 & 2 & 1 & 3 & -2 \\ 1 & -2 & 0 & 2 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$= [a_1 \ a_2 \ a_3 \ a_4 \ a_5]$$

Here, basic solutions comes from picking any pair of columns of A that forms an invertible 2×2 basis B , setting the other variables to 0.

Thus, excluding $[a_1, a_2]$ pair for its linear dependence, we examine the rest 9 pairs.

① Basis $[a_1, a_3]$	$B^{-1}b = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$	$(x_1, x_2, x_3, x_4, x_5) = (2, 0, 6, 0, 0)$
② Basis $[a_1, a_4]$	$B^{-1}b = \begin{bmatrix} -2/5 & 3/5 \\ 1/5 & 1/5 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} -2/5 \\ 6/5 \end{bmatrix}$	$(x_1, x_2, x_3, x_4, x_5) = (-2/5, 0, 0, 6/5, 0)$
③ Basis $[a_1, a_5]$	$B^{-1}b = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ -6 \end{bmatrix}$	$(x_1, x_2, x_3, x_4, x_5) = (8, 0, 0, 0, -6)$
④ Basis $[a_2, a_3]$	$B^{-1}b = \begin{bmatrix} 0 & -1/2 \\ 1 & 1/2 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 6 \end{bmatrix}$	$(x_1, x_2, x_3, x_4, x_5) = (0, -1, 6, 0, 0)$
⑤ Basis $[a_2, a_4]$	$B^{-1}b = \begin{bmatrix} 1/5 & -3/10 \\ 1/5 & 1/5 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 1/5 \\ 6/5 \end{bmatrix}$	$(x_1, x_2, x_3, x_4, x_5) = (0, 1/5, 0, 6/5, 0)$
⑥ Basis $[a_2, a_5]$	$B^{-1}b = \begin{bmatrix} -1/2 & -1 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 \\ -6 \end{bmatrix}$	$(x_1, x_2, x_3, x_4, x_5) = (0, -4, 0, 0, -6)$
⑦ Basis $[a_3, a_4]$	$B^{-1}b = \begin{bmatrix} 1 & -3/2 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$(x_1, x_2, x_3, x_4, x_5) = (0, 0, 1, 1, 0)$
⑧ Basis $[a_3, a_5]$	$B^{-1}b = \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$	$(x_1, x_2, x_3, x_4, x_5) = (0, 0, 8, 0, 2)$
⑨ Basis $[a_4, a_5]$	$B^{-1}b = \begin{bmatrix} 1/2 & 3/2 \\ -2/2 & 1/2 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 8/2 \\ -2/2 \end{bmatrix}$	$(x_1, x_2, x_3, x_4, x_5) = (0, 0, 0, 8/2, -2/2)$

#6 [2.25] Determine whether the following system possesses: (a) no solution, (b) a unique solution, or (c) many (how many?) solutions.

$$\begin{aligned} x_1 + 3x_2 + x_3 - x_4 &= 1 \\ 5x_2 - 6x_3 + x_4 &= 0 \\ x_1 - 2x_2 + 4x_3 &= 2. \end{aligned}$$

$$\left[\begin{array}{cccc|c} 1 & 3 & 1 & -1 & 1 \\ 0 & 5 & -6 & 1 & 0 \\ 1 & -2 & 4 & 0 & 2 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 3 & 1 & -1 & 1 \\ 0 & 5 & -6 & 1 & 0 \\ 0 & -5 & 3 & 1 & 1 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 3 & 1 & -1 & 1 \\ 0 & 5 & -6 & 1 & 0 \\ 0 & 0 & -3 & 2 & 1 \end{array} \right] : \text{rank} = 3$$

number of unknowns = 4. $\therefore$ degree of freedom = $4 - 3 = 1$.

Let $x_3 = t$, where t is an arbitrary real number

Then $x_4 = (3t + 1)/2$

$$x_2 = (9t - 1)/10$$

$$x_1 = (9 - 11t)/5$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 9/5 \\ -1/10 \\ 0 \\ 1/2 \end{bmatrix} + \begin{bmatrix} -11/5 \\ 9/10 \\ 1 \\ 3/2 \end{bmatrix} t \quad (t \in \mathbb{R})$$

$\therefore$ This system possesses infinitely many solutions.

7

[2.28] Which of the following sets are convex and which are not?

- $\{(x_1, x_2) : x_1^2 + x_2^2 \geq 3\}$.
- $\{(x_1, x_2, x_3) : x_1 + 2x_2 \leq 1, x_1 - 2x_3 \leq 2\}$.
- $\{(x_1, x_2) : x_2 - 3x_1^2 = 0\}$.
- $\{(x_1, x_2, x_3) : x_2 \geq x_1^2, x_1 + 2x_2 + x_3 \leq 4\}$.
- $\{(x_1, x_2) : x_1 = 3, |x_2| \leq 4\}$.
- $\{(x_1, x_2, x_3) : x_3 = |x_2|, x_1 \leq 3\}$.

a Not convex

 $a = (\sqrt{3}, 0)$, $b = (-\sqrt{3}, 0)$ are both included in the set.Let $c = \frac{1}{2}a + \frac{1}{2}b = (0, 0)$.However, since $0^2 + 0^2 = 0 \leq 3$, c is not included in the set. $\rightarrow$ NOT convexb Let $x = (x_1, x_2, x_3)$ and $y = (y_1, y_2, y_3)$ be points included in the set.

$$x_1 + 2x_2 \leq 1, \quad x_1 - 2x_3 \leq 2 \quad \text{holds.}$$

$$y_1 + 2y_2 \leq 1, \quad y_1 - 2y_3 \leq 2$$

Let $z = \theta x + (1-\theta)y$ ($0 \leq \theta \leq 1$)

$$\textcircled{1} z_1 + 2z_2$$

$$= (\theta x_1 + (1-\theta)y_1) + 2(\theta x_2 + (1-\theta)y_2)$$

$$= \theta(x_1 + 2x_2) + (1-\theta)(y_1 + 2y_2) \leq 1$$

$$\textcircled{2} z_1 - 2z_3$$

$$= (\theta x_1 + (1-\theta)y_1) - 2(\theta x_3 + (1-\theta)y_3)$$

$$= \theta(x_1 - 2x_3) + (1-\theta)(y_1 - 2y_3) \leq 2$$

z satisfies both conditions;
thus any segment connecting x and y
are included in the set.

 $\therefore$ convex

c Not convex

 $a = (-1, 3)$, $b = (1, 3)$ are both included in the set.Let $c = \frac{1}{2}a + \frac{1}{2}b = (0, 3)$.However, since $3 - 3 \cdot 0^2 = 3 \neq 0$, c is not included in the set. $\rightarrow$ NOT convexd Let $x = (x_1, x_2, x_3)$ and $y = (y_1, y_2, y_3)$ be points included in the set.Let $z = \theta x + (1-\theta)y$ ($0 \leq \theta \leq 1$)

$$\textcircled{1} x_2 \geq x_1^2, y_2 \geq y_1^2 \text{ holds.}$$

$$z_2 = \theta x_2 + (1-\theta)y_2 \geq \theta x_1^2 + (1-\theta)y_1^2$$

$$\geq (\theta x_1 + (1-\theta)y_1)^2$$

$$= z_1^2$$

$$\textcircled{2} x_1 + 2x_2 + x_3 \leq 4, y_1 + 2y_2 + y_3 \leq 4.$$

$$z_1 + 2z_2 + z_3 = \theta(x_1 + 2x_2 + x_3) + (1-\theta)(y_1 + 2y_2 + y_3) \leq 4$$

z satisfies both conditions;
thus any segment connecting x and y
are included in the set.

 $\therefore$ convexe Let $x = (3, x_2)$, $y = (3, y_2)$ be two points included in the set.

$$-4 \leq x_2 \leq 4 \quad \text{holds.}$$

$$-4 \leq y_2 \leq 4$$

Let $z = \theta x + (1-\theta)y$ ($0 \leq \theta \leq 1$)

$$\textcircled{1} z$$

$$= (3\theta + 3(1-\theta), \theta x_2 + (1-\theta)y_2)$$

$$= (3, \theta x_2 + (1-\theta)y_2) \rightarrow z_1 = 3$$

$$\textcircled{2} z_2 = \theta x_2 + (1-\theta)y_2 \leq 4\theta + 4(1-\theta) = 4$$

$$\geq -4\theta - 4(1-\theta) = -4$$

$$-4 \leq z_2 \leq 4. \rightarrow |z_2| \leq 4.$$

 $\therefore$ convex

f Not convex

 $a = (1, 2, 2)$, $b = (1, 2, 2)$ are both included in the set.Let $c = \frac{1}{2}a + \frac{1}{2}b = (1, 0, 2)$. $|c| \leq 3$ holds; yet, $2 \neq |0|$. Thus, c is not included in the set. $\rightarrow$ NOT convex

8

[2.39] Show that if C is a convex cone, then C has at most one extreme point, namely, the origin.

Let C be a convex cone where $x \in C, \lambda \geq 0 \rightarrow \lambda x \in C$

Recall that $x \in C$ is an extreme point if $x = \lambda x_1 + (1-\lambda)x_2, x_1, x_2 \in C, 0 < \lambda < 1 \rightarrow x = x_1 = x_2$

Assume there exists an extreme point $x \in C$ with $x \neq 0$.

Because C is a convex cone, ① $\lambda = 0 \rightarrow 0 \in C$

② $\lambda = 2 \rightarrow 2x \in C$

Taking $\lambda = \frac{1}{2}$, x can be written as a convex combination

$$x = \frac{1}{2} \cdot (0) + \frac{1}{2} \cdot (2x)$$

For x to become an extreme point, $x = 0 = 2x$ must hold.

Thus $x = 0 \rightarrow$ This contradicts our assumption that $x \neq 0$.

$\therefore$ If C is a convex cone, then C has at most one extreme point, the origin (0).

9

[2.40] Find the extreme points of the region defined by the following inequalities:

$$\begin{aligned} x_1 + 2x_2 + x_3 &\leq 5 \\ -x_1 + x_2 + 2x_3 &\leq 6 \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

(Hint: Consider $n = 3$ intersecting defining hyperplanes at a time.)

$$\text{let } x_1 + 2x_2 + x_3 = 5 \rightarrow \textcircled{A}$$

$$-x_1 + x_2 + 2x_3 = 6 \rightarrow \textcircled{B}$$

$$x_1 = 0 \rightarrow \textcircled{C}$$

$$x_2 = 0 \rightarrow \textcircled{D}$$

$$x_3 = 0 \rightarrow \textcircled{E}$$

1) A, B, C

$$x_1 = 0, x_2 = 4/3, x_3 = 7/3$$

2) A, B, D

$$x_1 = 4/3, x_2 = 0, x_3 = 11/3$$

3) A, B, E

$$x_1 = -7/3, x_2 = 11/3, x_3 = 0 \rightarrow (C \text{ violation})$$

4) A, C, D

$$x_1 = 0, x_2 = 0, x_3 = 5 \rightarrow (B \text{ violation})$$

5) A, C, E

$$x_1 = 0, x_2 = 5/2, x_3 = 0$$

6) A, D, E

$$x_1 = 5, x_2 = 0, x_3 = 0$$

7) B, C, D

$$x_1 = 0, x_2 = 0, x_3 = 3$$

8) B, C, E

$$x_1 = 0, x_2 = 6, x_3 = 0 \rightarrow (A \text{ violation})$$

9) B, D, E

$$x_1 = -6, x_2 = 0, x_3 = 0 \rightarrow (C \text{ violation})$$

10) C, D, E

$$x_1 = 0, x_2 = 0, x_3 = 0$$

#10

[2.41] Why does the following set have directions? Find all its extreme directions:

$$\begin{aligned} -x_1 + 2x_2 &= 3 \\ 2x_1 - 2x_2 - x_3 &\leq 2 \\ x_3 &\geq 1 \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

(Hint: Enumerate the extreme points of its normalized set of directions. What else needs to be checked?)

The following set has directions as x_3 can be increased unboundedly within the set's feasible region.

The recession cone of X : $D = \{d : -d_1 + 2d_2 = 0, 2d_1 - 2d_2 - d_3 \leq 0, d_1, d_2, d_3 \geq 0\}$
 $\hookrightarrow d_1 = 2d_2 \rightarrow 2d_2 - d_3 \leq 0$
 $= \{d : d_1 = 2d_2, d_3 \geq 2d_2, d_1, d_2, d_3 \geq 0\}$

Using the normalization constraint $D' = \{d \in D, d_1 + d_2 + d_3 = 1\}$
 $\hookrightarrow 3d_2 + d_3 = 1$

① $3d_2 + d_3 = 1, d_1 = 2d_2, d_1 = 0 \rightarrow (d_1, d_2, d_3) = (0, 0, 1)$

② $3d_2 + d_3 = 1, d_1 = 2d_2, d_2 = 0 \rightarrow \text{"} = (0, 0, 1)$

③ $3d_2 + d_3 = 1, d_1 = 2d_2, d_3 = 0 \rightarrow \text{"} = (\frac{2}{3}, \frac{1}{3}, 0) \rightarrow \text{violates } 2d_2 - d_3 \leq 0.$

④ $3d_2 + d_3 = 1, d_1 = 2d_2, 2d_2 - d_3 = 0 \rightarrow \text{"} = (\frac{2}{3}, \frac{1}{3}, \frac{2}{3})$

$\therefore \text{E.D.} : (0, 0, 1), (\frac{2}{3}, \frac{1}{3}, \frac{2}{3})$

[2.48] Let $X = \{(x_1, x_2) : x_1 - x_2 \leq 3, -x_1 + 3x_2 \leq 3, x_1 \geq -3\}$. Find all extreme points of X and represent $x = (0, 1)$ as a convex combination of the extreme points.

1) ① $x_1 - x_2 = 3$ & ② $-x_1 + 3x_2 = 3$

$\rightarrow x_1 = 6, x_2 = 3$

2) ① $x_1 - x_2 = 3$ & ③ $x_1 = -3$

$\rightarrow x_1 = -3, x_2 = -6$

3) ① $-x_1 + 3x_2 = 3$ & ③ $x_1 = -3$

$\rightarrow x_1 = -3, x_2 = 0$

}

Extreme Points of X

$: (-3, -6), (-3, 0), (6, 3)$

↓

Find $\alpha, \beta, \gamma \geq 0, \alpha + \beta + \gamma = 1$ that expresses

$\alpha(-3, -6) + \beta(-3, 0) + \gamma(6, 3) = (0, 1)$

$\rightarrow \alpha = 0, \beta = \frac{2}{3}, \gamma = \frac{1}{3}$

$\therefore (0, 1) = \frac{2}{3}(-3, 0) + \frac{1}{3}(6, 3)$

(2.1) $X = \{x : Ax \leq b, x \geq 0\}$, (2.2) $D = \{d : Ad \leq 0, 1^T d = 1, d \geq 0\}$

#12 [2.49] Answer the following questions and provide a brief explanation or illustration:

- Is it possible for X in Equation (2.1) to be empty but D in Equation (2.2) to be nonempty?
- Is there a relationship between redundancy and degeneracy of a polyhedral set?
- Does degeneracy imply redundancy in two dimensions?
- If the intersection of a finite number of half-spaces is nonempty, then this set has at least one extreme point. True or false? Explain.
- An unbounded n -dimensional polyhedral set can have at most n extreme directions. True or false? Explain.
- What is the maximum (actual) dimension of $X = \{x : Ax = b, x \geq 0\}$, where A is $m \times n$ of rank r , with $r \leq m \leq n$?

a Yes.

ex) Consider $X = \{x : Ax \leq b, x \geq 0\}$ where $A = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$, $b = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

For X , we need $x_1 \leq 0$ & $x_1 \geq 1 \rightarrow$ empty set. $X = \emptyset$

For D , $D = \{d : Ad \leq 0, d \geq 0, 1^T d = 1\}$, we can take $d = (0, 1)$. $D \neq \emptyset$

b Redundancy can cause degeneracy; while degeneracy does not always imply redundancy
 $\hookrightarrow$ If a redundant constraint is added to a fulfilled vertex, the vertex becomes degenerate.
 $\hookrightarrow$ A removal of a constraint intersecting a degenerate point can change the feasible region.

c No.

$A : \{x, y : x \geq 0, y \geq 0, x + y \leq 0\} \rightarrow$ Feasible region = $(0, 0)$

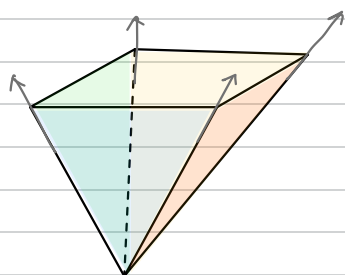
$\downarrow$
degenerate point.

Yet, the removal of either 3 constraints changes the feasible region.

d False

If the intersection results to be a line, an extreme point may not exist at all.

e False



: Example of a 3 dimensional unbounded polyhedron set having 4 extreme directions

f $n-r$

When the number of unknowns are n and $Ax=b$ enforces r variables, the maximum dimension is $n-r$.